

★ Discrete logarithm

The group $G = \langle \mathbb{Z}_p^*, x \rangle$ has several properties

1. Its elements include all integers from 1 to $p-1$
2. It always has primitive roots
3. It is cyclic
4. The primitive roots can be thought as the base of logarithm. If group has k primitive roots, calculation can be done in k different bases.
$$y = g^x \bmod n$$
$$\therefore x = \log_g y$$

→ for any element y in the set there is another element x that is the log of y in base g .

→ This type of logarithm is called discrete logarithm

→ we use notation $\log_g$ to show that base is g

★ Solution to modular logarithm using Discrete logs

How to solve problems of type $y = a^x \pmod n$ when y is given and we need to find x

Tabulation of discrete logarithms

→ This type of table can be precalculated and saved

y	1	2	3	4	5	6
$x = \log_3 y$	6	2	1	4	5	3

$x = \log_5 y$	6	4	5	2	1	3
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Discrete logarithm for $a \in \mathbb{Z}_7^*$

$$3^1 = 3^1 \pmod 7 = 3$$

$$3^2 = 2$$

$$3^3 = 6$$

$$3^4 = 4$$

$$3^5 = 5$$

$$3^6 = 3^6 \pmod 7 = 1$$

$$3^7 = 3^7 \pmod 7 = 3$$

$$5^1 = 5 \pmod 7 = 5$$

$$5^2 = 25 \pmod 7 = 4$$

$$5^3 = 6$$

$$5^4 = 2$$

$$5^5 = 3$$

$$5^6 = 1$$

Find x in each of following cases

① $4 \equiv 3^x \pmod{7}$

1. $x = \log_3 4 \pmod{7}$

$x = 4 \pmod{7}$

$3^x = 4$

$\log 3^x = \log 4$

$x \log 3 = \log 4$

$x = \frac{\log 4}{\log 3}$

$x = \log_3 4$

$x = \log_3 4$

② $6 \equiv 5^x \pmod{7}$

$x = \log_5 6 \pmod{7}$

$x = 3 \pmod{7}$

→ Tabulation and properties of discrete logarithms cannot be used to solve $y = a^x \pmod{n}$ when n is very large. Several algorithms have been devised that use the basic idea of discrete logarithm to solve the problem.